SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1708**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I __M Kavya 192472370___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: kavya

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Task 1: Article Summary

1(a) Citation and Domain

Citation: Zhao, L.; Lee, S.; Jeong, S.-P. Decision Tree Application to Classification Problems with Boosting Algorithm. Electronics 2021, 10(16), 1903.

DOI: https://doi.org/10.3390/electronics10161903

Publisher / Source: MDPI, Electronics journal (Special Issue: Advances in Machine Learning), peer-reviewed, open access.

Domain / Application Area: Financial technology / retail banking – personal credit and customer-behaviour classification.

ML Problem Addressed: A binary classification problem that predicts whether a bank customer will subscribe to a fixed (term) deposit, based on demographic and campaign-contact attributes, using decision tree models strengthened with a boosting algorithm.

1(b) Summary (150–200 words)

The article investigates whether combining classical decision tree algorithms with a boosting technique can improve customer-classification accuracy for a bank marketing dataset of over 45,000 records. The authors build two baseline models, C5.0 and Classification and Regression Tree (CART), and then enhance each with boosting and, for C5.0, an additional cost matrix to handle class imbalance. The research gap identified is that earlier decision tree studies on imbalanced financial data often report inflated overall accuracy while badly under-predicting the minority class of customers who actually subscribe to a deposit, and few studies compare boosting's effect on C5.0 against CART on the same dataset. The authors propose to fill this gap by systematically evaluating accuracy, precision, sensitivity, specificity, F1-score, Kappa value, and model-fitting time across five model variants, validated with 10-fold cross-validation, so that the trade-offs of boosting can be judged beyond raw accuracy alone.

Task 2: Methodology Analysis

2(a) ML Technique and Dataset

The study uses two decision tree variants: the C5.0 algorithm, which splits nodes using information gain ratio, and the CART algorithm, which splits nodes using the Gini coefficient. Both are combined with the AdaBoost-style boosting procedure, where successive weak classifiers are trained on re-weighted versions of the training data and combined into a single strong classifier. For C5.0, a cost matrix is additionally introduced so that misclassifying a genuine subscriber is penalised more heavily than misclassifying a non-subscriber, directly addressing class imbalance.

The dataset is the well-known Bank of Portugal marketing dataset, containing 45,211 customer records (40,690 used for model fitting after preprocessing, with a 4,521-record hold-out test set). It has 16 input attributes — 7 continuous variables such as age, account balance, call duration and number of prior contacts, and 9 discrete variables such as job type, marital status, education, and housing-loan status — plus a binary target variable indicating whether the customer subscribed to a term deposit. Before model fitting, the authors checked Pearson correlation and Variance Inflation Factor (VIF) among the continuous variables; all correlations were below 0.5 and all VIF values were under 2, confirming no serious multicollinearity, so no dimensionality reduction was required.

2(b) Mapping to CO4 and Strength/Limitation

This methodology maps directly onto the CO4 topic of decision tree algorithms: C5.0 and CART are classical inductive-learning, tree-based classifiers, and the boosting procedure is a widely used ensemble extension of decision tree learning covered under statistical/ensemble learning methods.

Methodological strength: The authors evaluate models using multiple complementary metrics (accuracy, precision, sensitivity, specificity, F1-score, Kappa) rather than accuracy alone, and further validate results with 10-fold cross-validation, which gives a much more balanced picture of performance on an imbalanced dataset than a single train/test split would.

Methodological limitation: The dataset is heavily imbalanced (far more non-subscribers than subscribers), and although a cost matrix is used for C5.0, the same class-imbalance correction is not applied to CART; this makes the cross-model comparison somewhat uneven and may understate how much CART's sensitivity could improve with similar rebalancing.

Task 3: Results and Critical Evaluation

3(a) Key Results

Across all five model variants, overall accuracy stayed close to 90%, but precision, sensitivity, and F1-score varied considerably. CART with boosting achieved the highest raw accuracy (90.95%) and the largest jump in precision after boosting was applied (from 37.24% to 65.23%), while C5.0 with the cost matrix achieved the best sensitivity and F1-score for identifying actual subscribers (70.32% precision, 61.29% F1-score), despite having the lowest raw accuracy of the five models. The comparative results reported by the authors are summarised below.

Model	Accuracy	Precision	Sensitivity	F1-Score	Fitting Time
C5.0	90.56%	45.60%	63.24%	53.05%	4 s
C5.0 + Cost Matrix	89.60%	70.32%	54.31%	61.29%	4 s
C5.0 + Boosting	90.33%	45.18%	61.92%	52.24%	8 s
CART	90.31%	37.24%	65.02%	47.36%	10 s
CART + Boosting	90.95%	65.23%	48.58%	55.69%	~2 min

3(b) Critical Evaluation of the Metrics

The reported metrics appear realistic rather than inflated: the authors do not stop at the headline ~90% accuracy figure, which on this imbalanced dataset would be achievable even by a trivial model that always predicts "no subscription." Instead, they foreground precision and sensitivity for the minority class, which honestly exposes CART's weak precision (37.24%) before boosting. A possible bias is that the correlation and VIF checks were performed only on the seven continuous variables, leaving the nine discrete/categorical variables (job, marital status, education, and others) unexamined for redundancy or encoding bias, and the 10-fold cross-validation figures in the paper are reported only for accuracy and Kappa, not for precision or sensitivity, which are the metrics that matter most for this imbalanced problem. Additional experiments that would strengthen the findings include reporting AUC-ROC curves for each model, applying resampling techniques such as SMOTE alongside the cost matrix, and testing statistical significance of the accuracy differences between models rather than comparing point estimates alone.

3(c) Real-World Applicability

This decision tree and boosting approach is directly deployable in retail-banking marketing analytics, where it can rank customers by likelihood of subscribing to term deposits so call-centre resources are focused on high-probability leads, and the same pipeline generalises to other propensity-to-buy or churn-prediction problems in telecom and insurance. Scaling this to industry practice raises several challenges: data availability and quality (the model depends on accurate, up-to-date contact-history logs that may be incomplete or inconsistent across branches), compute cost (the paper notes that CART with boosting alone took around two minutes to fit versus a few seconds for the unboosted models, a gap that grows substantially at production data volumes), and ethical concerns around using demographic attributes such as marital status or education in a credit- or offer-targeting model, which can risk indirect discrimination and would need fairness auditing before deployment.

Task 4: Reflection and Learning Outcome

4(a) Three Key Insights

1. Boosting does not uniformly improve every base learner: the article shows that boosting improved CART substantially (higher precision and F1-score) but barely improved C5.0, because C5.0 already reaches its most Kappa-favourable configuration at a single training round (trials = 1). This connects directly to the CO4 concept of ensemble/statistical learning methods, and shows that ensemble techniques must be evaluated empirically rather than assumed to always help.

2. Accuracy alone is a misleading metric on imbalanced classification problems. The C5.0 + cost matrix model had the lowest overall accuracy of the five variants but the best sensitivity and F1-score for the minority (subscriber) class, reinforcing the course concept that classification model evaluation requires a confusion-matrix-based reading (precision, sensitivity, specificity, Kappa) rather than accuracy in isolation.

3. Decision tree algorithms make different, inspectable structural choices depending on their splitting criterion — C5.0 (information gain ratio) used nearly all 16 attributes and produced a tree of size 414, while CART (Gini coefficient) used only 2 attributes and produced a much smaller tree of size 74, yet reached comparable accuracy. This illustrates the inductive-learning principle from CO4 that different splitting heuristics can arrive at very different, but similarly effective, hypotheses from the same data.

4(b) Proposed Extension

A worthwhile extension would be to re-run the same C5.0/CART/boosting pipeline on a resampled version of the dataset (e.g., using SMOTE to oversample the subscriber class) combined with the existing cost matrix, and to apply the same experiment to a different domain, such as telecom customer-churn data, to test whether the sensitivity gains from cost-sensitive boosting generalise beyond banking. This is feasible because the SMOTE technique and gradient boosting libraries used in the study are freely available in standard R and Python machine-learning packages, and churn datasets of comparable size are publicly available (e.g., the IBM Telco Customer Churn dataset). The expected impact is a more balanced classifier that keeps the interpretability of a decision tree while substantially improving minority-class sensitivity without the roughly 2-minute fitting-time penalty that boosting alone incurred on CART in the original study, since SMOTE operates on the data rather than adding sequential boosting rounds.

REFERENCES:

[1] L. Zhao, S. Lee, and S.-P. Jeong, "Decision tree application to classification problems with boosting algorithm," *Electronics*, vol. 10, no. 16, p. 1903, Aug. 2021, doi: 10.3390/electronics10161903.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand